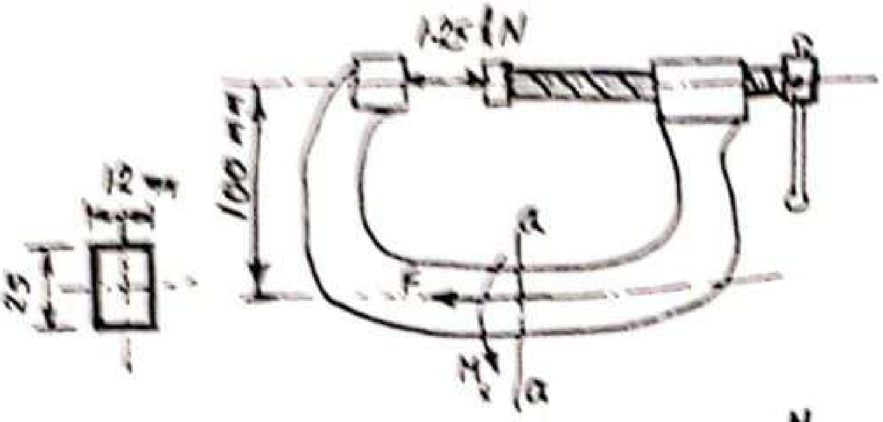
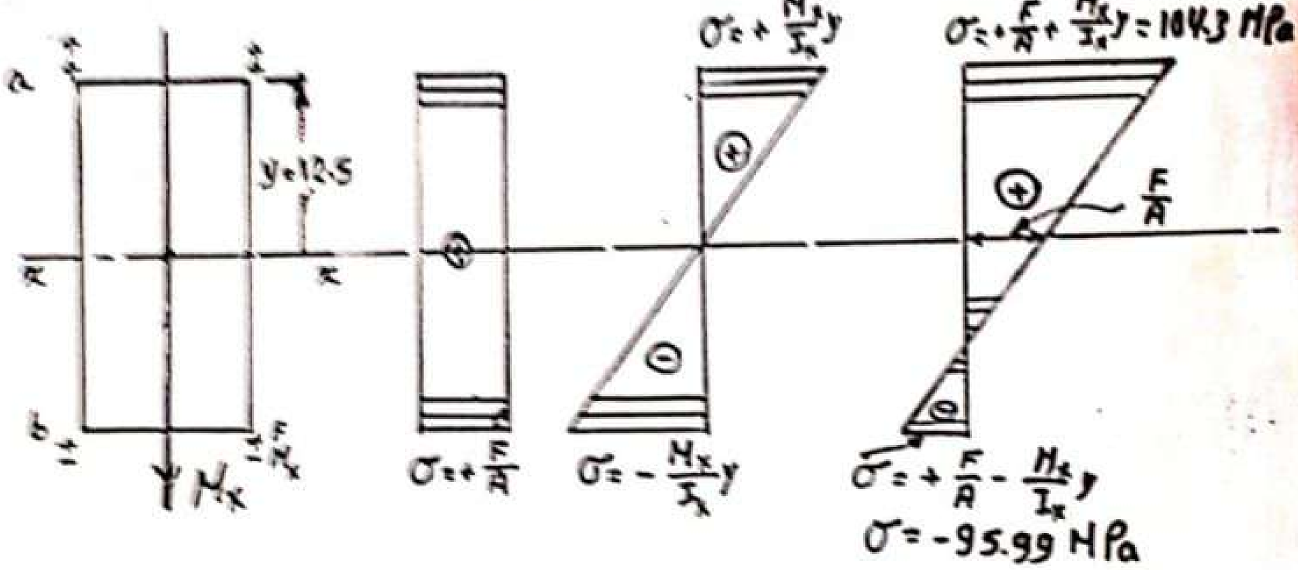
Find the maximum stresses on section a-a of the shown clamp when the force provided by the screw is 1.25 kN. Plot the stress distribution across the section.





$A = 12 \times 25 = 300 \ mm^2$

$I_x = \frac{12(25)^3}{12} = 15.6 \times 10^3 \ mm^4$

$F = 1.25 \times 10^3 \ N$, $M_x = 1.25 \times 10^3 \times 10^2 = 125 \times 10^3 \ Nmm$

$$\sigma_{a,b} = \frac{F}{A} \pm \frac{M_x}{I_x} y = \frac{1.25 \times 10^3 \ N}{300 \ mm^2} \pm \frac{125 \times 10^3 (N.mm) \times 12.5 \ mm}{15.6 \times 10^3 \ mm^4}$$

$\sigma_a = 104.3 \ N/mm^2$

$\sigma_b = -95.99 \ N/mm^2$

$\boxed{\sigma_a = 104.3 \ MPa \quad \sigma_b = -95.99 \ MPa}$

$\boxed{1 \ N/mm^2 = 1 \ MN/m^2 = 1 \ MPa}$